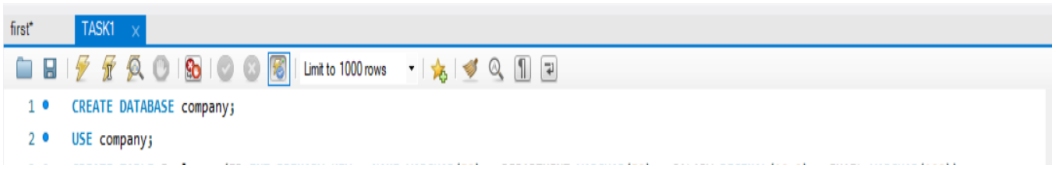
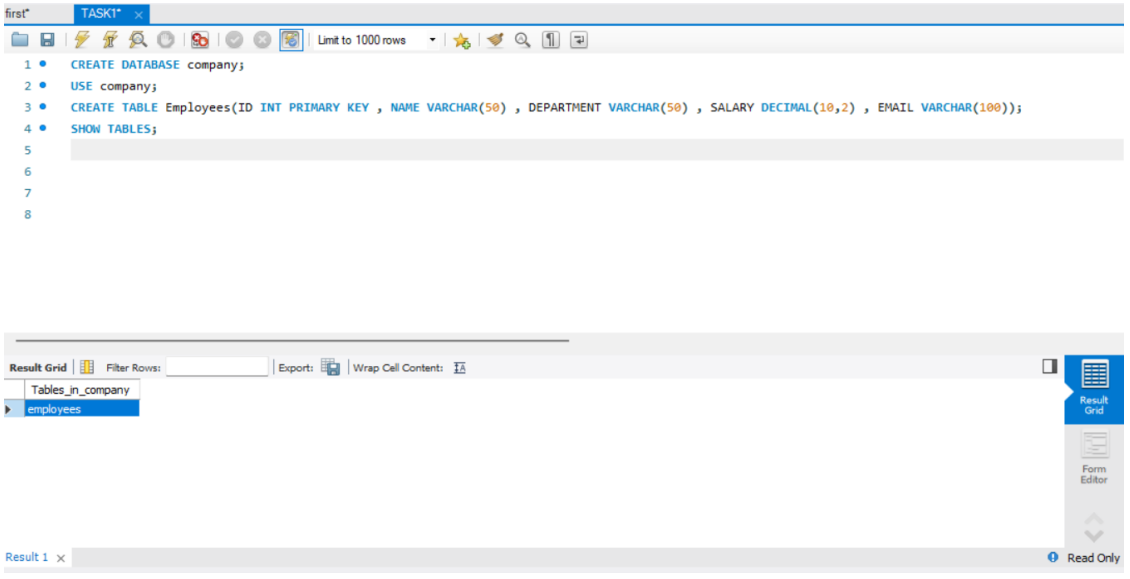
Task 1: Create a Table→

A) **Create a database named Company:**

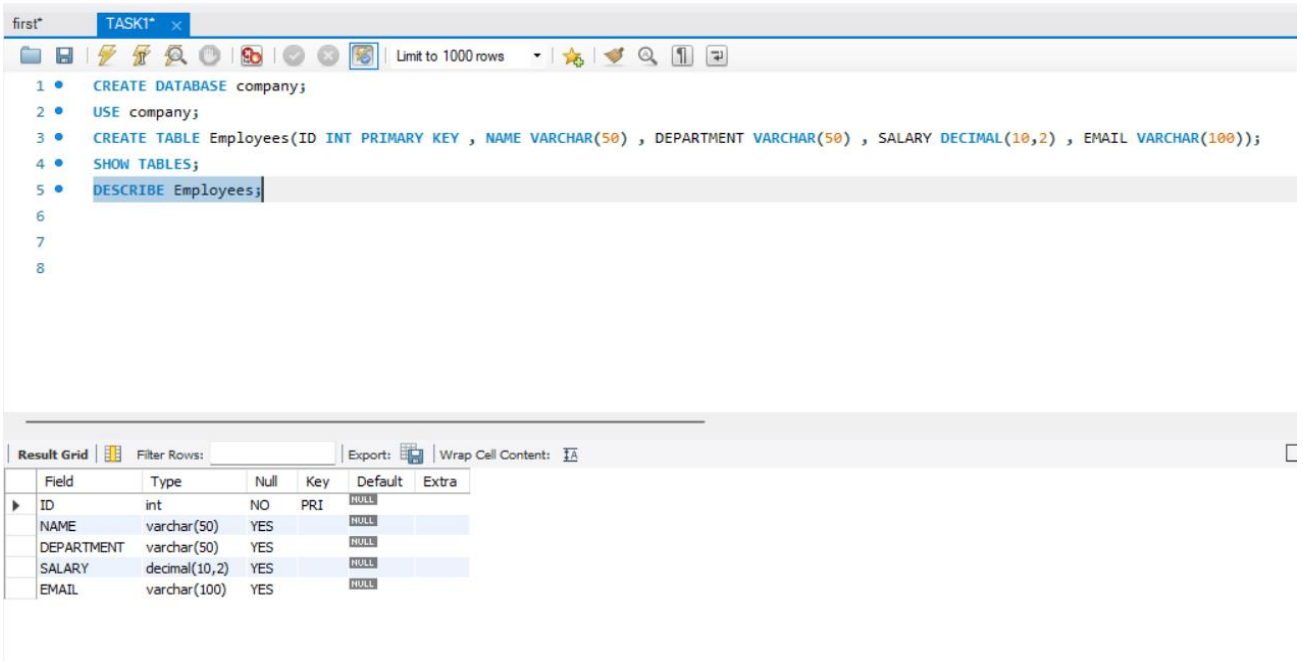


B) Create a table named Employees :

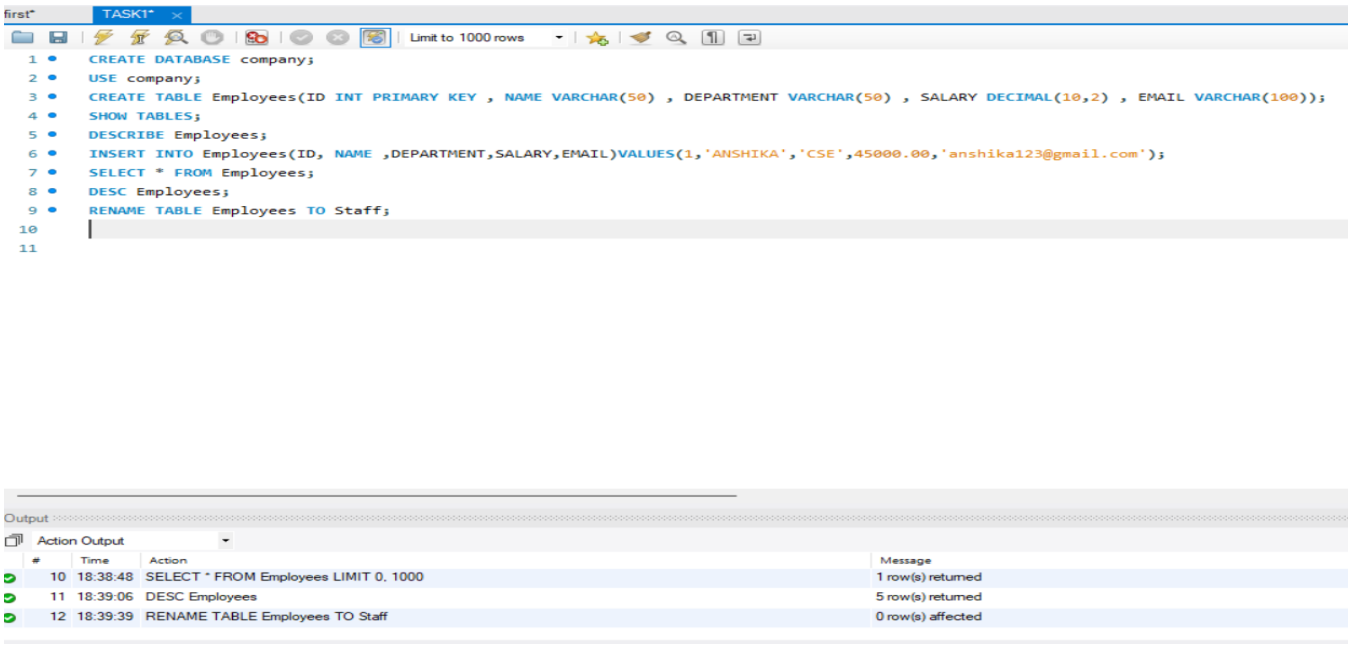


Task 2: DDL Queries on the Employees Table→

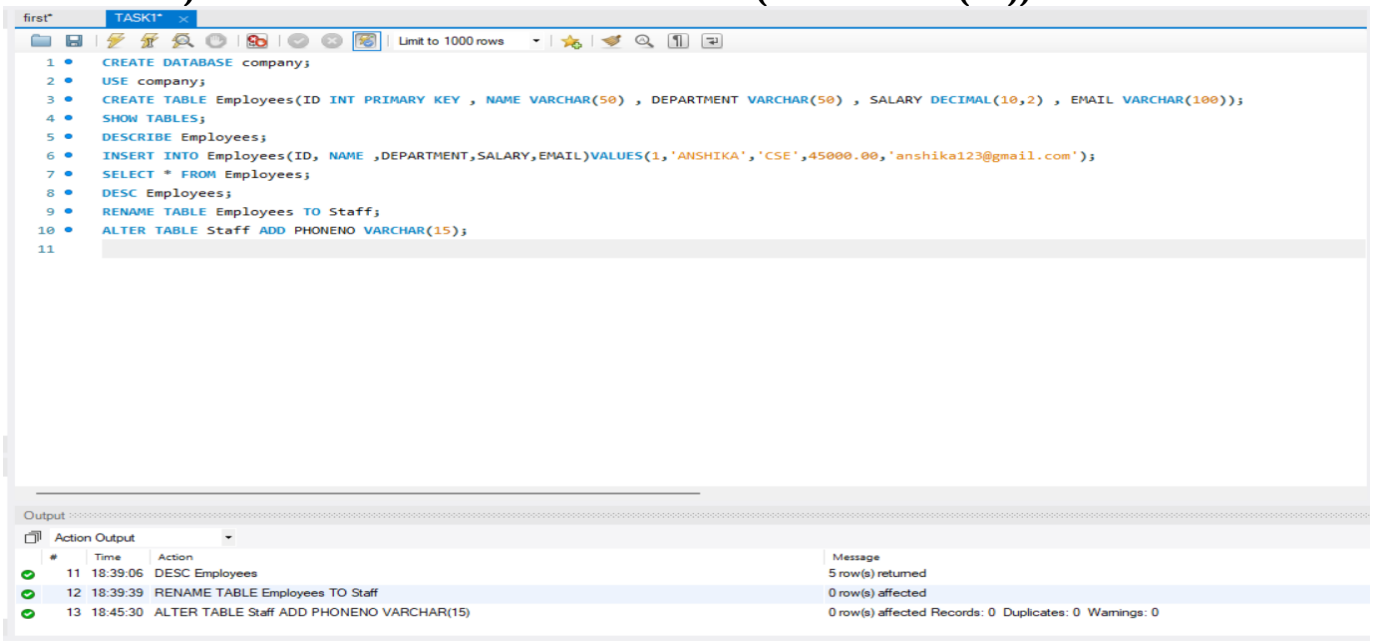
A) **Describe the structure of the table:**



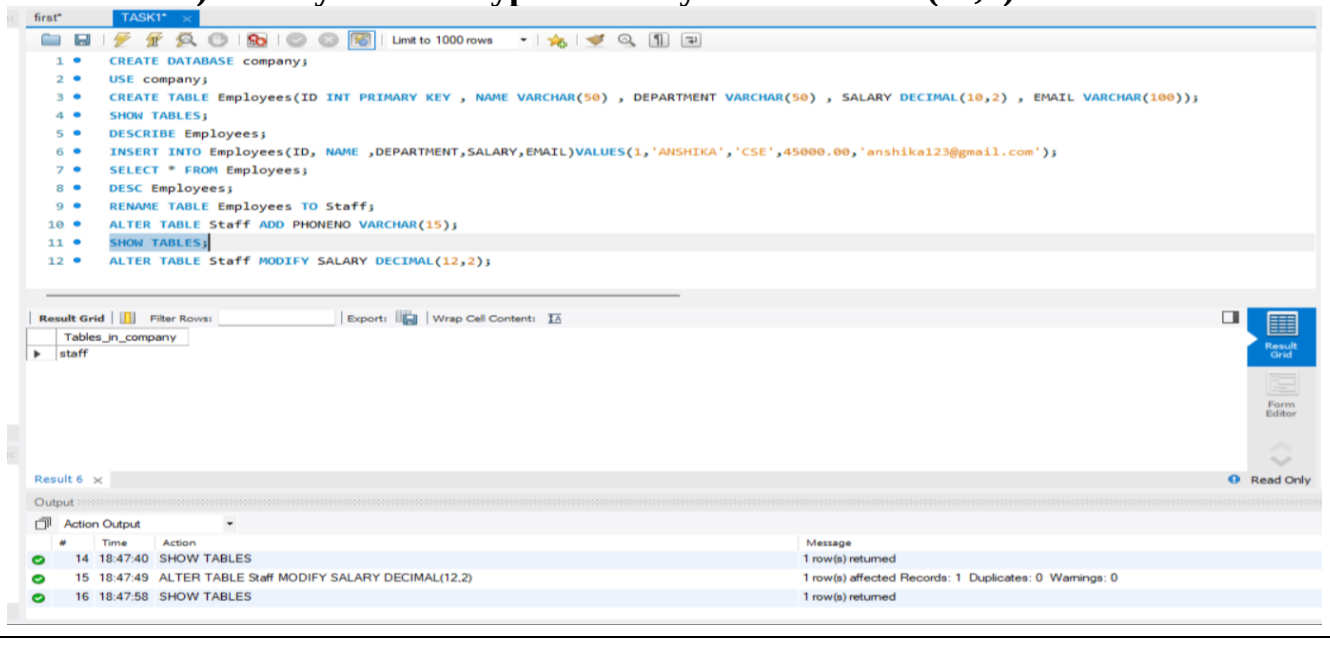
B) **Rename the table to Staff:**



C) Add a new column Phone Number (VARCHAR(15)):



D) Modify the data type of Salary to DECIMAL(12,2):



E)**Add a DEFAULT value of 'General' to the Department column:**

```
12 • ALTER TABLE Staff MODIFY SALARY DECIMAL(12,2);
13 • ALTER TABLE Staff MODIFY DEPARTMENT VARCHAR(50) DEFAULT 'GENERAL';
```

```
11 • SHOW TABLES;
12 • ALTER TABLE Staff MODIFY SALARY DECIMAL(12,2);
13 • ALTER TABLE Staff MODIFY DEPARTMENT VARCHAR(50) DEFAULT 'GENERAL';
14 • INSERT INTO Staff(ID,NAME,SALARY,EMAIL)VALUES(2,'GAURI',45000.00,'gauri163@gmail,com'),(3,'priya',65000.00,'priya163@gmail,com'),(4,'SUMAN',10(
15 • SELECT * FROM Staff;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

ID	NAME	DEPARTMENT	SALARY	EMAIL	PHONENO
1	ANSHIKA	CSE	45000.00	anshika123@gmail.com	NULL
2	GAURI	NULL	45000.00	gauri163@gmail,com	NULL
3	priya	NULL	65000.00	priya163@gmail,com	NULL
4	SUMAN	NULL	10000.00	suman173@gmail,com	NULL
NULL	NULL	NULL	NULL	NULL	NULL

Import records from an external file

Result Grid | Form Editor

Staff 1 | Apply | Revert

Output

Action Output

#	Time	Action	Message
3	19:53:07	LTER TABLE Staff MODIFY DEPARTMENT VARCHAR(50) DEFAULT 'GENERAL'	Error Code: 1064. You have an error in your SQL syntax; check the manual that correspon
4	19:56:40	INSERT INTO Staff(ID,NAME,SALARY,EMAIL)VALUES(2,'GAURI',45000.00,'gauri163@gmail,com'),(3,'priya',6...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
5	19:56:53	SELECT * FROM Staff LIMIT 0, 1000	4 row(s) returned

F)**Drop the Phone Number column:**

```
15 • SELECT * FROM Staff;
16 • ALTER TABLE Staff DROP COLUMN PHONENO;
```

Administration | Schemas

Information

No object selected

Object Info | Session

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	19:56:40	INSERT INTO Staff(ID,NAME,SALARY,EMAIL)VALUES(2,'GAURI',45000.00,'gauri163@gmail,com'),(3,'priya',6...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
5	19:56:53	SELECT * FROM Staff LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
6	20:07:29	ALTER TABLE Staff DROP COLUMN PHONENO	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.062 sec

```
12 • ALTER TABLE Staff MODIFY SALARY DECIMAL(12,2);
13 • ALTER TABLE Staff MODIFY DEPARTMENT VARCHAR(50) DEFAULT 'GENERAL';
14 • INSERT INTO Staff(ID,NAME,SALARY,EMAIL)VALUES(2,'GAURI',45000.00,'gauri163@gmail,com'),(3,'priya',65000.00,'priya163@gmail,com'),(4,'SUMAN',10(
15 • SELECT * FROM Staff;
16 • ALTER TABLE Staff DROP COLUMN PHONENO;
17 • SHOW TABLES;
18 • SELECT * FROM Staff;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

ID	NAME	DEPARTMENT	SALARY	EMAIL
1	ANSHIKA	CSE	45000.00	anshika123@gmail.com
2	GAURI	NULL	45000.00	gauri163@gmail,com
3	priya	NULL	65000.00	priya163@gmail,com
4	SUMAN	NULL	10000.00	suman173@gmail,com
NULL	NULL	NULL	NULL	NULL

Result Grid

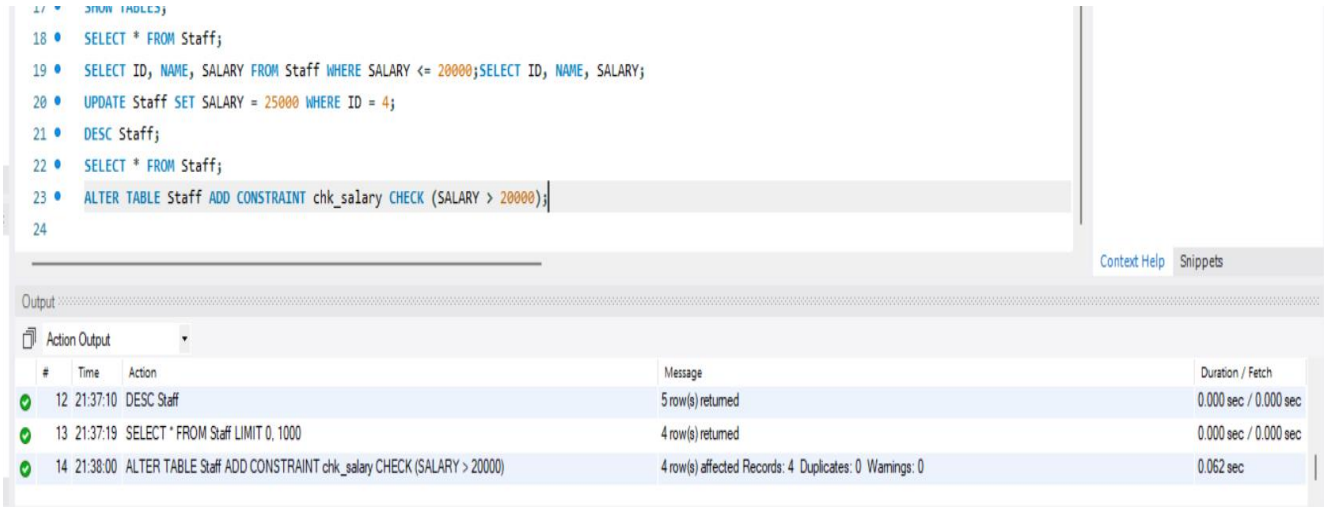
Staff 3 | Apply | Revert | Context Help | Snippets

Output

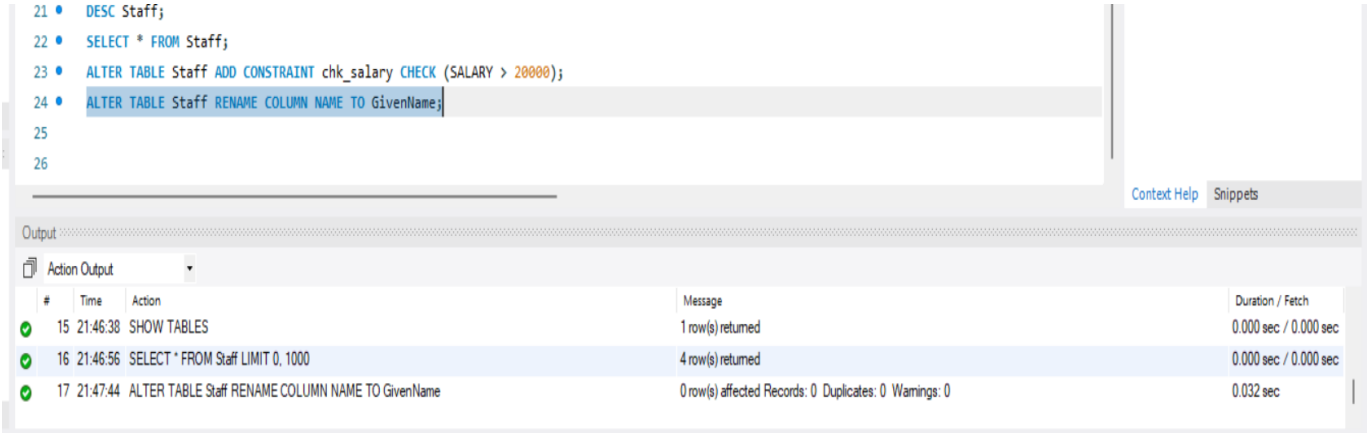
Action Output

#	Time	Action	Message	Duration / Fetch
6	20:07:29	ALTER TABLE Staff DROP COLUMN PHONENO	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.062 sec
7	20:11:28	SHOW TABLES	1 row(s) returned	0.000 sec / 0.000 sec
8	20:12:27	SELECT * FROM Staff LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

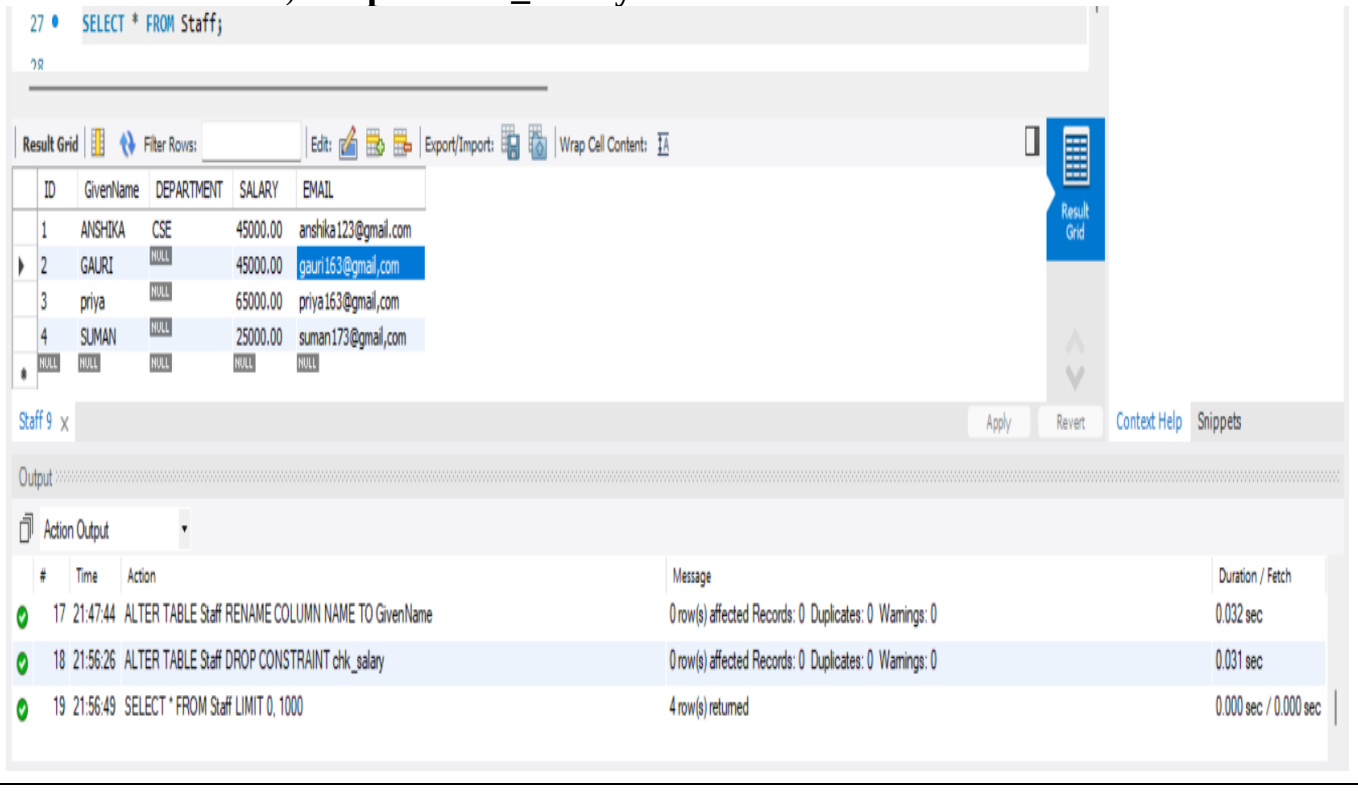
G) Add a CHECK constraint to ensure Salary is greater than 20000:



H) Rename the column FirstName to GivenName:



I) Drop the chk_Salary constraint:



J) Create an index on the Email column:

```
SHOW CREATE TABLE Staff;
CREATE INDEX idx_email ON Staff(EMAIL);
SHOW INDEX FROM Staff;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
staff	0	PRIMARY	1	ID	A	4	NULL	NULL		BTREE			YES	NULL
staff	1	idx_email	1	EMAIL	A	4	NULL	NULL	YES	BTREE			YES	NULL

Result 10 — Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
19	21:56:49	SELECT * FROM Staff LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
20	22:09:38	CREATE INDEX idx_email ON Staff(EMAIL)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.032 sec
21	22:12:12	SHOW INDEX FROM Staff	2 row(s) returned	0.016 sec / 0.000 sec

K) Drop the index on the Email column:

```
SHOW INDEX FROM Staff;
DROP INDEX idx_email ON Staff;
SHOW INDEX FROM Staff;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
staff	0	PRIMARY	1	ID	A	4	NULL	NULL		BTREE			YES	NULL

Result 11 — Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
21	22:12:12	SHOW INDEX FROM Staff	2 row(s) returned	0.016 sec / 0.000 sec
22	22:14:16	DROP INDEX idx_email ON Staff	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
23	22:14:21	SHOW INDEX FROM Staff	1 row(s) returned	0.000 sec / 0.000 sec

L) Truncate the table to remove all rows without affecting the structure:

```
SHOW INDEX FROM Staff;
TRUNCATE TABLE Staff;
SELECT * FROM Staff;
DESC Staff;
```

ID	GivenName	DEPARTMENT	SALARY	EMAIL
NULL	NULL	NULL	NULL	NULL

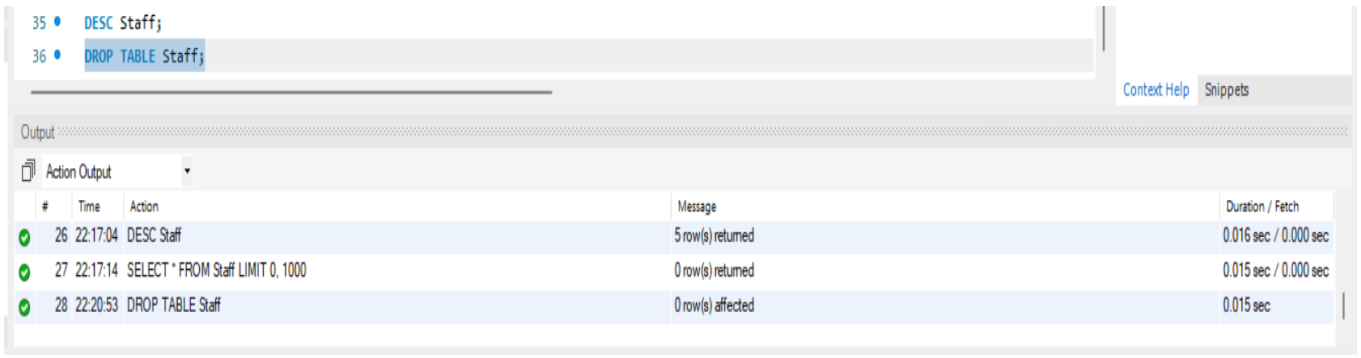
Staff 14 — Apply — Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
25	22:16:59	SELECT * FROM Staff LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
26	22:17:04	DESC Staff	5 row(s) returned	0.016 sec / 0.000 sec
27	22:17:14	SELECT * FROM Staff LIMIT 0, 1000	0 row(s) returned	0.015 sec / 0.000 sec

M) **Drop the Staff table completely:**



3.Conceptual Questions:

1. **What is the difference between TRUNCATE and DELETE?**

ANS-	**DELETE**	**TRUNCATE**
	Deletes selected rows or all rows from a table.	Deletes **all rows** from a table.
	Can use WHERE condition.	Cannot use WHERE.
	Example: DELETE FROM Student WHERE id=5;	Example: TRUNCATE TABLE Student;
	Slower for deleting all records because rows are deleted individually.	Faster because it removes all records at once.
	Can be rolled back in many DBMSs when used within a transaction.	Rollback behavior depends on the DBMS.
	Table structure remains.	Table structure remains.
	DELETE is a **DML** command.	TRUNCATE is generally classified as **DDL**.

2. **Explain the use of the CHECK constraint.**

ANS- The CHECK constraint is used to ensure that the values entered into a table satisfy a specific condition.

It prevents invalid data from being inserted into the table.

Syntax:

```
CREATE TABLE Student (
   id INT,
   age INT CHECK (age >= 18)
);
```

Here, CHECK (age >= 18) means that the student's age must be **18 or above**.

3. **Why is it important to define a PRIMARY KEY for a table?**

ANS- A PRIMARY KEY is important because it uniquely identifies each record (row) in a table.

Main points:

It does not allow duplicate values.

It does not allow NULL values.

It helps identify each record uniquely.

.

4. **How does ALTER TABLE help in modifying an existing table?**

ANS- ALTER TABLE is used to modify the structure of an existing table without deleting the table.

It can be used to:

1. **Add a new column**

   ALTER TABLE Student ADD age INT;

2. **Drop (remove) a column**

   ALTER TABLE Student DROP COLUMN age;

3. **Modify a column**

   ALTER TABLE Student MODIFY name VARCHAR(100);

5. **What is the purpose of creating an index in a table?**

ANS- An INDEX is created to make data retrieval/searching faster in a table.

For example, if a table has thousands of student records, an index on student_id helps the database find a particular student more quickly.

Example:

CREATE INDEX idx_student_name

ON Student(name);

This creates an index on the name column**.**

4.Exercises:

1. **Create a table named Projects with the following structure:**



2. Perform at least 5 DDL operations on the Projects table.

```
CREATE TABLE Projects (
    ProjectID INT PRIMARY KEY AUTO_INCREMENT,
    ProjectName VARCHAR(100) NOT NULL,
    StartDate DATE,
    EndDate DATE,
    Budget DECIMAL(15,2) CHECK (Budget > 0)
);
ALTER TABLE Projects
ADD COLUMN ManagerName VARCHAR(100);

ALTER TABLE Projects
MODIFY COLUMN ProjectName VARCHAR(150) NOT NULL;

ALTER TABLE Projects
RENAME COLUMN ManagerName TO ProjectManager;

ALTER TABLE Projects
DROP COLUMN ProjectManager;

CREATE INDEX idx_ProjectName
ON Projects(ProjectName);

DESC Projects;
```

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Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
32	22:55:56	ALTER TABLE Projects MODIFY COLUMN ProjectName VARCHAR(150) NOT NULL	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec
33	22:56:04	ALTER TABLE Projects RENAME COLUMN ManagerName TO ProjectManager	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.032 sec
34	22:56:09	ALTER TABLE Projects DROP COLUMN ProjectManager	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
35	22:56:14	CREATE INDEX idx_ProjectName ON Projects(ProjectName)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec

A) Write a query to retrieve the names of all columns in the Staff table.

```
DESC Projects;
SELECT COLUMN_NAME
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_SCHEMA = DATABASE()
AND TABLE_NAME = 'Staff';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

COLUMN_NAME

Result Grid

COLUMNS 3 x

Read Only | Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
38	22:58:58	USE company	0 row(s) affected	0.000 sec
39	22:59:02	CREATE TABLE Projects (ProjectID INT PRIMARY KEY AUTO_INCREMENT, ProjectName VARCHAR(...	Error Code: 1050. Table 'projects' already exists	0.000 sec
40	23:06:03	SELECT COLUMN_NAME FROM INFORMATION_SCHEMA.COLUMNS WHERE TABLE_SCHEMA = DATA...	0 row(s) returned	0.015 sec / 0.000 sec

B) Write a query to modify the Department column to allow NULL values

